

Mathematics for Data Science

Lecture 1 - Recap and Proofs

Vector Spaces, Subspaces, Linear Maps, Rank and Kernel

This document gathers the proofs of the main propositions and theorems from Lecture 1. It is intended both as a lecture recap and as an exercise sheet: each proof is an exercise :) Some proofs are deliberately written as explanation, rather than in a purely abstract manner with quantifiers. Whatever your way of writing them, always abide by the principle that all notations must be defined! When writing sums, always precise the range $\sum_{min}^{max}$.

NB: the order is slightly different than the order of appearance in the slides. Some basic facts about bases and dimension are proved first, because several later results use them. You can recognize several structure types for the proofs, such as (i) Existence and Uniqueness; (ii) Verifying the properties for a given object, (iii) Induction. (iv) Proving equivalence by proving one implication ($\implies$), then the other ($\impliedby$), etc.

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1 Bases and dimension

1.1 Unique coordinates in a basis

Proposition

Given a basis, every vector has a unique coordinate representation.

Proof. Let V be a vector space and let $\mathcal{B} = (e_1, \dots, e_n)$ be a basis of V .

Existence. Since $\mathcal{B}$ is a basis, it is in particular a spanning family of V . Therefore, for every $x \in V$, there exist scalars $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that $x = \alpha_1 e_1 + \dots + \alpha_n e_n$.

This proves that every vector has at least one coordinate representation in the basis $\mathcal{B}$.

Uniqueness. Assume that x has two coordinate representations in the basis $\mathcal{B}$. Then there exist scalars $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$ such that $x = \alpha_1 e_1 + \dots + \alpha_n e_n$ and $x = \beta_1 e_1 + \dots + \beta_n e_n$.

Subtracting these two equalities gives $0 = (\alpha_1 - \beta_1)e_1 + \dots + (\alpha_n - \beta_n)e_n$.

Since $\mathcal{B} = (e_1, \dots, e_n)$ is a basis, it is linearly independent. Therefore all coefficients in the previous linear combination must be zero. Hence $\alpha_i - \beta_i = 0$ for every $i = 1, \dots, n$, so $\alpha_i = \beta_i$ for every i .

Thus the coordinate representation is unique. $\square$

1.2 The exchange lemma

Lemma – Exchange lemma

Let V be a vector space. Let $(v_1, \dots, v_p)$ be a linearly independent family in V , and let $(w_1, \dots, w_q)$ be a spanning family of V . Then $p \leq q$.
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Proof. The idea is the following: starting from the spanning family $(w_1, \dots, w_q)$, we can replace some of the w_j by the vectors v_i one after another, while keeping a spanning family. Since there are only q vectors in the spanning family, we cannot replace more than q of them. Therefore $p \leq q$.

We give the argument more explicitly.

Since $(w_1, \dots, w_q)$ spans V , the vector v_1 can be written as a linear combination of the w_j : $v_1 = a_1 w_1 + \dots + a_q w_q$. Since $v_1 \neq 0$, at least one coefficient a_j is nonzero. We may assume, after reordering the w_j if necessary, that $a_1 \neq 0$.

Then we can solve this equality for w_1 : $w_1 = \frac{1}{a_1} v_1 - \frac{a_2}{a_1} w_2 - \dots - \frac{a_q}{a_1} w_q$.

This shows that every vector that could be generated using $w_1, w_2, \dots, w_q$ can also be generated using $v_1, w_2, \dots, w_q$. Thus $(v_1, w_2, \dots, w_q)$ is still a spanning family of V .

Now we repeat the same idea with v_2 . Since $(v_1, w_2, \dots, w_q)$ spans V , the vector v_2 can be written as a linear combination of $v_1, w_2, \dots, w_q$. The vector v_2 cannot be a multiple of v_1 only, because (v_1, v_2) is linearly independent. Therefore at least one of the vectors $w_2, \dots, w_q$ must have a nonzero coefficient in this expression. We can then replace one of these w_j by v_2 and keep a spanning family.

Continuing in this way, after inserting $v_1, \dots, v_r$, we still have a spanning family containing these r vectors. This process can continue only as long as there remain vectors among the original $w_1, \dots, w_q$ to replace. Therefore we must have $p \leq q$. $\square$

1.3 Finite-dimensional vector spaces

Proposition

Let V be a finite-dimensional vector space. Then:

- (1) V admits a basis.
- (2) All bases of V have the same cardinality.
- (3) Any family of linearly independent vectors from V can be completed to obtain a basis of V .
- (4) A basis of V can be extracted from any spanning family of V .

Proof. We prove the four statements separately.

1. *Existence of a basis.* Since V is finite-dimensional, there exists a finite spanning family $(v_1, \dots, v_m)$ of V .

If this family is linearly independent, then it is both spanning and linearly independent, so it is a basis.

If it is linearly dependent, then one of the vectors can be written as a linear combination of the others. Removing this vector does not change the span of the family. Indeed, any linear combination that used this vector can be rewritten using the other vectors.

We repeat this operation as long as the family is linearly dependent. Since the initial family is finite, this process stops after finitely many steps. At the end, we have a family that is still spanning and is now linearly independent. Therefore it is a basis of V .

2. *All bases have the same cardinality.* Let $\mathcal{B} = (e_1, \dots, e_p)$ and $\mathcal{C} = (f_1, \dots, f_q)$ be two bases of V .

Since $\mathcal{B}$ is linearly independent and $\mathcal{C}$ spans V , the exchange lemma gives $p \leq q$.

Conversely, since $\mathcal{C}$ is linearly independent and $\mathcal{B}$ spans V , the exchange lemma gives $q \leq p$.

Therefore $p = q$. Thus all bases of V have the same cardinality.

3. *Completing a linearly independent family.* Let $(u_1, \dots, u_r)$ be a linearly independent family in V .

If this family spans V , then it is a basis and there is nothing to prove.

Otherwise, there exists a vector $v \in V$ such that $v \notin \text{span}(u_1, \dots, u_r)$. We claim that $(u_1, \dots, u_r, v)$ is linearly independent.

Indeed, suppose that $a_1 u_1 + \dots + a_r u_r + a v = 0$. If $a \neq 0$, then $v = -\frac{a_1}{a} u_1 - \dots - \frac{a_r}{a} u_r$, which would imply $v \in \text{span}(u_1, \dots, u_r)$, contradiction. Hence $a = 0$. Then $a_1 u_1 + \dots + a_r u_r = 0$, and since $(u_1, \dots, u_r)$ is linearly independent, all $a_i = 0$.

Thus we can add v and keep a linearly independent family. Since V is finite-dimensional, this process cannot continue forever. Eventually, we obtain a linearly independent family that spans V , hence a basis.

4. *Extracting a basis from a spanning family.* Let $(w_1, \dots, w_s)$ be a spanning family of V .

If it is linearly independent, it is already a basis.

If it is linearly dependent, one vector is a linear combination of the others, and we can remove it without changing the span. We repeat this operation until the remaining family is linearly independent. Since the family is finite, the process stops.

The final family is still spanning and is now linearly independent. Therefore it is a basis extracted from the original spanning family. $\square$

2 Vector spaces and product spaces

2.1 Product of vector spaces

Proposition

Endowed with the above operations, $(E \times F, +, \cdot)$ is an $\mathbb{F}$ -vector space. It is called the product vector space of E by F .

Proof. Let E and F be two vector spaces over $\mathbb{F}$. We define the operations on $E \times F$ by $(x, y) + (x', y') = (x + x', y + y')$ and $\lambda(x, y) = (\lambda x, \lambda y)$.

To prove that $E \times F$ is a vector space, we verify the vector space axioms.

Closure. Let $(x, y), (x', y') \in E \times F$. Since E is a vector space, $x + x' \in E$. Since F is a vector space, $y + y' \in F$. Hence $(x + x', y + y') \in E \times F$.

Similarly, if $\lambda \in \mathbb{F}$ and $(x, y) \in E \times F$, then $\lambda x \in E$ and $\lambda y \in F$, so $\lambda(x, y) \in E \times F$.

Zero vector and opposites. The zero vector is $(0_E, 0_F)$, because for every $(x, y) \in E \times F$, $(x, y) + (0_E, 0_F) = (x + 0_E, y + 0_F) = (x, y)$.

The opposite of (x, y) is $(-x, -y)$, because $(x, y) + (-x, -y) = (0_E, 0_F)$.

Other axioms. Associativity and commutativity of addition hold componentwise because they hold in E and in F . For example, $((x, y) + (x', y')) + (x'', y'') = ((x + x') + x'', (y + y') + y'')$, which is equal to $(x + (x' + x''), y + (y' + y''))$ by associativity in E and F .

The distributive properties and compatibility of scalar multiplication also hold componentwise. For instance, $\lambda((x, y) + (x', y')) = \lambda(x + x', y + y') = (\lambda x + \lambda x', \lambda y + \lambda y') = \lambda(x, y) + \lambda(x', y')$.

Thus all vector space axioms are satisfied, and $E \times F$ is a vector space over $\mathbb{F}$. $\square$

Proposition

Let $n > 1$ be an integer, and suppose that $E_1, \dots, E_n$ are finite-dimensional vector spaces over $\mathbb{F}$. Then their Cartesian product is finite-dimensional, with dimension

$$\dim(E_1 \times E_2 \times \cdots \times E_n) = \sum_{i=1}^n \dim(E_i).$$

Proof. We first prove the result for two vector spaces E and F .

Let $(e_1, \dots, e_p)$ be a basis of E and $(f_1, \dots, f_q)$ be a basis of F . We claim that the following family is a basis of $E \times F$:

$$(e_1, 0_F), \dots, (e_p, 0_F), (0_E, f_1), \dots, (0_E, f_q).$$

Spanning property. Let $(x, y) \in E \times F$. Since $(e_1, \dots, e_p)$ is a basis of E , there exist scalars $\alpha_1, \dots, \alpha_p$ such that $x = \alpha_1 e_1 + \cdots + \alpha_p e_p$. Similarly, since $(f_1, \dots, f_q)$ is a basis of F , there exist scalars $\beta_1, \dots, \beta_q$ such that $y = \beta_1 f_1 + \cdots + \beta_q f_q$.

Therefore $(x, y) = \alpha_1(e_1, 0_F) + \cdots + \alpha_p(e_p, 0_F) + \beta_1(0_E, f_1) + \cdots + \beta_q(0_E, f_q)$. So the family spans $E \times F$.

Linear independence. Assume that $\alpha_1(e_1, 0_F) + \cdots + \alpha_p(e_p, 0_F) + \beta_1(0_E, f_1) + \cdots + \beta_q(0_E, f_q) = (0_E, 0_F)$.

Then $(\alpha_1 e_1 + \cdots + \alpha_p e_p, \beta_1 f_1 + \cdots + \beta_q f_q) = (0_E, 0_F)$.

Hence $\alpha_1 e_1 + \cdots + \alpha_p e_p = 0_E$ and $\beta_1 f_1 + \cdots + \beta_q f_q = 0_F$. Since both bases are linearly independent, all α_i and all β_j are zero.

So the family is linearly independent.

Thus it is a basis of $E \times F$, and it has $p+q$ elements. Hence $\dim(E \times F) = \dim(E) + \dim(F)$. The general formula for $E_1 \times \cdots \times E_n$ follows by applying the two-space case repeatedly.

- **Initialization.** For $n = 2$, we have proven the property.
- **Heredity.** Suppose the property holds for a given integer $n \geq 2$. We want to prove that it holds for $n + 1$.
To this aim, we remark that $E' = E_1 \times E_2 \times \cdots \times E_n$ is itself a vector space (this is the previous property). Applying the case $n = 2$ to $E' \times E_{n+1}$, we write

$$\dim(E' \times E_{n+1}) = \dim(E') + \dim(E_{n+1}) = \sum_{i=1}^{n+1} \dim(E_i).$$

- We conclude that the property holds for *any* integer $n > 1$.

□

3 Subspaces

3.1 Subspaces are vector spaces

Proposition

Let V be a vector space over $\mathbb{F}$. Any subspace U of V is a vector space over $\mathbb{F}$.

Proof. The operations on U are the same as the operations on V , restricted to elements of U .

Since U is a subspace of V , we know that $0_V \in U$, that U is stable under addition, and that U is stable under scalar multiplication.

All algebraic identities required in the definition of a vector space already hold in V . For instance, for $u_1, u_2, u_3 \in U$, the equality $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$ holds because these vectors are also elements of V .

The zero vector of U is 0_V , which belongs to U . If $u \in U$, then $-u = (-1)u \in U$ because U is stable under scalar multiplication.

Thus U satisfies all vector space axioms over $\mathbb{F}$. Therefore U is a vector space over $\mathbb{F}$. □

3.2 Intersection of subspaces

Proposition

Let F and G be subspaces of a vector space E . Then $F \cap G$ is a subspace of E .

Proof. We use the subspace criterion: we prove that $F \cap G$ contains the zero vector, is stable under addition, and is stable under scalar multiplication.

First, since F and G are subspaces of E , we have $0_E \in F$ and $0_E \in G$. Hence $0_E \in F \cap G$.

Let $x, y \in F \cap G$. Then $x, y \in F$ and $x, y \in G$. Since F is a subspace, $x + y \in F$. Since G is a subspace, $x + y \in G$. Therefore $x + y \in F \cap G$.

Let $x \in F \cap G$ and $\lambda \in \mathbb{F}$. Then $x \in F$ and $x \in G$. Since F and G are subspaces, $\lambda x \in F$ and $\lambda x \in G$. Therefore $\lambda x \in F \cap G$.

Thus $F \cap G$ is a subspace of E . □

3.3 A homogeneous linear equation defines a subspace

Proposition

Let $n \in \mathbb{N}$ and $a_1, a_2, \dots, a_n \in \mathbb{R}$. The following set F is a vector subspace of $\mathbb{R}^n$:

$$F = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid \sum_{i=1}^n a_i x_i = 0\}.$$

Proof. We again use the subspace criterion.

First, $0 = (0, \dots, 0)$ belongs to F , because $\sum_{i=1}^n a_i 0 = 0$.

Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ be elements of F . Then $\sum_{i=1}^n a_i x_i = 0$ and $\sum_{i=1}^n a_i y_i = 0$. We compute $\sum_{i=1}^n a_i (x_i + y_i) = \sum_{i=1}^n a_i x_i + \sum_{i=1}^n a_i y_i = 0 + 0 = 0$. Hence $x + y \in F$.

Let $x = (x_1, \dots, x_n) \in F$ and $\lambda \in \mathbb{R}$. Then $\sum_{i=1}^n a_i (\lambda x_i) = \lambda \sum_{i=1}^n a_i x_i = \lambda \cdot 0 = 0$. Hence $\lambda x \in F$.

Therefore F is a vector subspace of $\mathbb{R}^n$. $\square$

4 Sum of subspaces

4.1 The sum of two subspaces

Auxiliary proposition

Let F and G be subspaces of a vector space E . Then

$$F + G = \{x \in E \mid \exists u \in F, \exists w \in G, x = u + w\}$$

is a subspace of E .

Proof.

First, $0_E = 0_E + 0_E$, with $0_E \in F$ and $0_E \in G$. Hence $0_E \in F + G$.

Let $x, y \in F + G$. Then there exist $u, u' \in F$ and $w, w' \in G$ such that $x = u + w$ and $y = u' + w'$. Therefore $x + y = (u + w) + (u' + w') = (u + u') + (w + w')$. Since F is a subspace, $u + u' \in F$. Since G is a subspace, $w + w' \in G$. Hence $x + y \in F + G$.

Let $x \in F + G$ and $\lambda \in \mathbb{F}$. Write $x = u + w$ with $u \in F$ and $w \in G$. Then $\lambda x = \lambda u + \lambda w$. Since $\lambda u \in F$ and $\lambda w \in G$, we get $\lambda x \in F + G$.

Thus $F + G$ is a subspace of E . $\square$

4.2 Direct sums and uniqueness

Proposition

If $F \cap G = \{0\}$, then every vector in $F + G$ can be written uniquely as the sum of a vector from F and a vector from G .

Proof. By definition of $F + G$, every vector $x \in F + G$ can be written at least once as $x = u + w$, with $u \in F$ and $w \in G$. So existence is immediate.

We prove uniqueness. Suppose that $x \in F + G$ has two decompositions: $x = u + w$ and $x = u' + w'$, where $u, u' \in F$ and $w, w' \in G$.

Then $u + w = u' + w'$, so $u - u' = w' - w$.

Now $u - u' \in F$, because F is a subspace. Also $w' - w \in G$, because G is a subspace. Since these two vectors are equal, this common vector belongs to $F \cap G$.

But $F \cap G = \{0\}$. Hence $u - u' = 0$ and $w' - w = 0$. Therefore $u = u'$ and $w = w'$.

So the decomposition is unique. $\square$

5 Linear maps

5.1 Kernel, range, injectivity and surjectivity

Proposition

If $L : U \rightarrow V$ is a linear map, then:

- (1) the kernel of L is a subspace of U ;
- (2) the range of L is a subspace of V ;
- (3) L is injective iff $\ker(L) = \{0_U\}$;
- (4) L is surjective iff $\text{range}(L) = V$.

Proof. We prove each statement separately.

1. *The kernel is a subspace.* Recall that $\ker(L) = \{x \in U \mid L(x) = 0_V\}$.

First, $0_U \in \ker(L)$ because $L(0_U) = 0_V$. Indeed, $L(0_U) = L(0_U + 0_U) = L(0_U) + L(0_U)$, so subtracting $L(0_U)$ from both sides gives $L(0_U) = 0_V$.

Let $x, y \in \ker(L)$. Then $L(x) = 0_V$ and $L(y) = 0_V$. By linearity, $L(x + y) = L(x) + L(y) = 0_V + 0_V = 0_V$. So $x + y \in \ker(L)$.

Let $x \in \ker(L)$ and $\lambda \in \mathbb{F}$. Then $L(\lambda x) = \lambda L(x) = \lambda 0_V = 0_V$. So $\lambda x \in \ker(L)$.

Therefore $\ker(L)$ is a subspace of U .

2. *The range is a subspace.* Recall that $\text{range}(L) = \{y \in V \mid \exists x \in U, y = L(x)\}$.

First, $0_V \in \text{range}(L)$ because $0_V = L(0_U)$.

Let $y_1, y_2 \in \text{range}(L)$. Then there exist $x_1, x_2 \in U$ such that $y_1 = L(x_1)$ and $y_2 = L(x_2)$. Then $y_1 + y_2 = L(x_1) + L(x_2) = L(x_1 + x_2)$. Thus $y_1 + y_2 \in \text{range}(L)$.

Let $y_1 \in \text{range}(L)$ and $\lambda \in \mathbb{F}$. Write $y_1 = L(x_1)$ for some $x_1 \in U$. Then $\lambda y_1 = \lambda L(x_1) = L(\lambda x_1)$. Thus $\lambda y_1 \in \text{range}(L)$.

Therefore $\text{range}(L)$ is a subspace of V .

3. *Injectivity and the kernel.* Assume first that L is injective. Let $x \in \ker(L)$. Then $L(x) = 0_V = L(0_U)$. Since L is injective, $x = 0_U$. Hence $\ker(L) = \{0_U\}$.

Conversely, assume that $\ker(L) = \{0_U\}$. Let $x, y \in U$ such that $L(x) = L(y)$. Then $L(x - y) = L(x) - L(y) = 0_V$. Thus $x - y \in \ker(L)$, so $x - y = 0_U$. Therefore $x = y$. Hence L is injective.

4. *Surjectivity and the range.* By definition, L is surjective from U to V iff every vector $v \in V$ is of the form $v = L(u)$ for some $u \in U$. This is exactly the statement $\text{range}(L) = V$. $\square$

5.2 Range of the zero map

Proposition – Range of the zero map

If $L : U \rightarrow V$ is a linear map, then

$$\text{range}(L) = \{0_V\} \iff L = 0.$$

Proof. We prove both implications.

Assume first that $L = 0$. This means that for every $x \in U$, $L(x) = 0_V$. Therefore the only vector reached by L is 0_V , so $\text{range}(L) = \{0_V\}$.

Conversely, assume that $\text{range}(L) = \{0_V\}$. Let $x \in U$. Since $L(x)$ belongs to $\text{range}(L)$, and since $\text{range}(L) = \{0_V\}$, we must have $L(x) = 0_V$.

This is true for every $x \in U$, so L is the zero map. $\square$

5.3 A linear map is determined by the image of a basis

Proposition

Let U, V be two vector spaces defined over $\mathbb{F}$, $\mathcal{B}_U = (u_1, u_2, \dots, u_n)$ a basis of U , and $\mathcal{F}_V = (v_1, v_2, \dots, v_n)$ a family of vectors in V . Then there exists a unique linear map $g : U \rightarrow V$ such that, for all i with $1 \leq i \leq n$, $g(u_i) = v_i$.

In addition:

- (1) g is injective iff $\mathcal{F}_V$ is linearly independent.
- (2) g is surjective iff $\mathcal{F}_V$ spans V .
- (3) g is bijective iff $\mathcal{F}_V$ is a basis of V .

Proof. We first construct the map, then prove uniqueness, then prove the three additional statements.

Existence. Let $x \in U$. Since $\mathcal{B}_U = (u_1, \dots, u_n)$ is a basis of U , there exist unique scalars $\alpha_1, \dots, \alpha_n$ such that $x = \alpha_1 u_1 + \dots + \alpha_n u_n$.

Define $g(x) = \alpha_1 v_1 + \dots + \alpha_n v_n$.

This definition makes sense because the coefficients α_i are uniquely determined by x .

We now check that g is linear. Let $x, y \in U$, with $x = \sum_{i=1}^n \alpha_i u_i$ and $y = \sum_{i=1}^n \beta_i u_i$. For $\lambda, \mu \in \mathbb{F}$, we have $\lambda x + \mu y = \sum_{i=1}^n (\lambda \alpha_i + \mu \beta_i) u_i$. Therefore $g(\lambda x + \mu y) = \sum_{i=1}^n (\lambda \alpha_i + \mu \beta_i) v_i = \lambda \sum_{i=1}^n \alpha_i v_i + \mu \sum_{i=1}^n \beta_i v_i = \lambda g(x) + \mu g(y)$. So g is linear.

Also, for each j , the coordinate representation of u_j in the basis $\mathcal{B}_U$ has coefficient 1 in position j and 0 elsewhere. Hence $g(u_j) = v_j$.

Uniqueness. Let $h : U \rightarrow V$ be another linear map such that $h(u_i) = v_i$ for all i .

For $x = \alpha_1 u_1 + \dots + \alpha_n u_n$, linearity gives $h(x) = \alpha_1 h(u_1) + \dots + \alpha_n h(u_n) = \alpha_1 v_1 + \dots + \alpha_n v_n = g(x)$.

Thus $h(x) = g(x)$ for every $x \in U$, so $h = g$. The map is unique.

Injectivity. Assume first that $(v_1, \dots, v_n)$ is linearly independent. Let $x \in \ker(g)$. Write $x = \alpha_1 u_1 + \dots + \alpha_n u_n$. Then $0_V = g(x) = \alpha_1 v_1 + \dots + \alpha_n v_n$. Since the v_i are linearly independent, all $\alpha_i = 0$. Hence $x = 0_U$. Thus $\ker(g) = \{0_U\}$, so g is injective.

Conversely, assume that $(v_1, \dots, v_n)$ is linearly dependent. Then there exist scalars $\alpha_1, \dots, \alpha_n$, not all zero, such that $\alpha_1 v_1 + \dots + \alpha_n v_n = 0_V$. Let $x = \alpha_1 u_1 + \dots + \alpha_n u_n$. Since the u_i are

linearly independent and the coefficients are not all zero, $x \neq 0_U$. But $g(x) = 0_V$. Hence $\ker(g) \neq \{0_U\}$, so g is not injective.

Therefore g is injective iff $(v_1, \dots, v_n)$ is linearly independent.

Surjectivity. We observe that $\text{range}(g) = \text{span}(v_1, \dots, v_n)$.

Indeed, every vector of the form $g(x)$ is a linear combination of $v_1, \dots, v_n$. Conversely, every linear combination $\alpha_1 v_1 + \dots + \alpha_n v_n$ is equal to $g(\alpha_1 u_1 + \dots + \alpha_n u_n)$.

Thus g is surjective iff $\text{range}(g) = V$, iff $\text{span}(v_1, \dots, v_n) = V$, i.e. iff $(v_1, \dots, v_n)$ spans V .

Bijectivity. The map g is bijective iff it is both injective and surjective. By the two previous parts, this is equivalent to saying that $(v_1, \dots, v_n)$ is both linearly independent and spanning. That is exactly the definition of being a basis of V . $\square$

5.4 Isomorphism of finite-dimensional vector spaces

Proposition

Finite-dimensional vector spaces defined over the same field and of the same dimension are isomorphic.

Proof. Let U and V be finite-dimensional vector spaces over the same field $\mathbb{F}$, and assume that $\dim(U) = \dim(V) = n$.

Let $\mathcal{B}_U = (u_1, \dots, u_n)$ be a basis of U , and let $\mathcal{B}_V = (v_1, \dots, v_n)$ be a basis of V .

By the previous proposition, there exists a unique linear map $g : U \rightarrow V$ such that $g(u_i) = v_i$ for every $i = 1, \dots, n$.

Since $(v_1, \dots, v_n)$ is a basis of V , it is linearly independent and spans V . By the previous proposition again, this implies that g is injective and surjective. Hence g is bijective.

Therefore g is an isomorphism from U to V . Thus U and V are isomorphic. $\square$

6 Rank

6.1 Rank-nullity theorem

Theorem – Rank-nullity theorem

If U, V are finite-dimensional vector spaces and $L : U \rightarrow V$ is linear, then

$$\text{rank}(L) + \dim(\ker(L)) = \dim(U).$$

Proof. Let $p = \dim(\ker(L))$. Choose a basis $(e_1, \dots, e_p)$ of $\ker(L)$.

Since $\ker(L)$ is a subspace of U , and U is finite-dimensional, we can complete this linearly independent family into a basis of U . Thus there exist vectors $u_1, \dots, u_q$ such that $(e_1, \dots, e_p, u_1, \dots, u_q)$ is a basis of U .

We will prove that $(L(u_1), \dots, L(u_q))$ is a basis of $\text{range}(L)$.

Spanning property. Let $y \in \text{range}(L)$. Then there exists $x \in U$ such that $y = L(x)$.

Since $(e_1, \dots, e_p, u_1, \dots, u_q)$ is a basis of U , there exist scalars α_i and β_j such that $x = \alpha_1 e_1 + \dots + \alpha_p e_p + \beta_1 u_1 + \dots + \beta_q u_q$.

Applying L , we get $y = L(x) = \alpha_1 L(e_1) + \dots + \alpha_p L(e_p) + \beta_1 L(u_1) + \dots + \beta_q L(u_q)$.

But each e_i belongs to $\ker(L)$, so $L(e_i) = 0_V$. Therefore $y = \beta_1 L(u_1) + \dots + \beta_q L(u_q)$.

Thus $(L(u_1), \dots, L(u_q))$ spans $\text{range}(L)$.

Linear independence. Assume that $\beta_1 L(u_1) + \dots + \beta_q L(u_q) = 0_V$.

Then $L(\beta_1 u_1 + \cdots + \beta_q u_q) = 0_V$. Thus $\beta_1 u_1 + \cdots + \beta_q u_q \in \ker(L)$.

Since $(e_1, \dots, e_p)$ is a basis of $\ker(L)$, there exist scalars $\alpha_1, \dots, \alpha_p$ such that $\beta_1 u_1 + \cdots + \beta_q u_q = \alpha_1 e_1 + \cdots + \alpha_p e_p$.

Rearranging gives $(-\alpha_1)e_1 + \cdots + (-\alpha_p)e_p + \beta_1 u_1 + \cdots + \beta_q u_q = 0_U$.

But $(e_1, \dots, e_p, u_1, \dots, u_q)$ is a basis of U , so it is linearly independent. Therefore all coefficients are zero. In particular, $\beta_1 = \cdots = \beta_q = 0$.

Thus $(L(u_1), \dots, L(u_q))$ is linearly independent.

Conclusion. We have proved that $(L(u_1), \dots, L(u_q))$ is a basis of $\text{range}(L)$. Therefore $\text{rank}(L) = q$.

On the other hand, $\dim(\ker(L)) = p$ and $\dim(U) = p + q$, because $(e_1, \dots, e_p, u_1, \dots, u_q)$ is a basis of U .

Hence $\text{rank}(L) + \dim(\ker(L)) = q + p = \dim(U)$. $\square$

6.2 Rank and injectivity/surjectivity

Proposition

Let $L : U \rightarrow V$ be a linear map, with U and V finite-dimensional. Then:

- (1) L is injective iff $\text{rank}(L) = \dim(U)$.
- (2) L is surjective iff $\text{rank}(L) = \dim(V)$.
- (3) L is bijective iff $\text{rank}(L) = \dim(U) = \dim(V)$.

Proof. We use two previously proved facts:

- L is injective iff $\ker(L) = \{0_U\}$;
- $\text{rank}(L) + \dim(\ker(L)) = \dim(U)$.

Injectivity. The map L is injective iff $\ker(L) = \{0_U\}$. This is equivalent to $\dim(\ker(L)) = 0$.

By rank-nullity, $\text{rank}(L) + \dim(\ker(L)) = \dim(U)$. Therefore $\dim(\ker(L)) = 0$ iff $\text{rank}(L) = \dim(U)$.

So L is injective iff $\text{rank}(L) = \dim(U)$.

Surjectivity. The map L is surjective iff $\text{range}(L) = V$. Since $\text{rank}(L) = \dim(\text{range}(L))$, this is equivalent to $\text{rank}(L) = \dim(V)$.

Bijectivity. The map L is bijective iff it is both injective and surjective. By the two previous parts, this is equivalent to $\text{rank}(L) = \dim(U)$ and $\text{rank}(L) = \dim(V)$.

Therefore L is bijective iff $\text{rank}(L) = \dim(U) = \dim(V)$. $\square$

7 Dimension formula for sums of subspaces

Proposition

For U and W finite-dimensional subspaces of a vector space V ,

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

Proof. We prove the result by introducing a linear map whose kernel is related to $U \cap W$.

Define $\varphi : U \times W \rightarrow U + W$ by $\varphi(u, w) = u + w$.

Linearity of φ . Let $(u, w), (u', w') \in U \times W$ and let $\lambda, \mu \in \mathbb{F}$. Then

$$\begin{aligned}\varphi(\lambda(u, w) + \mu(u', w')) &= \varphi(\lambda u + \mu u', \lambda w + \mu w') \\ &= (\lambda u + \mu u') + (\lambda w + \mu w') \\ &= \lambda(u + w) + \mu(u' + w') \\ &= \lambda\varphi(u, w) + \mu\varphi(u', w').\end{aligned}$$

So φ is linear.

Range of φ . By definition, every element of $U + W$ is of the form $u + w$ with $u \in U$ and $w \in W$. Therefore $\text{range}(\varphi) = U + W$. Hence $\text{rank}(\varphi) = \dim(U + W)$.

Kernel of φ . We have $(u, w) \in \ker(\varphi)$ iff $\varphi(u, w) = 0$, iff $u + w = 0$, iff $w = -u$.

Since $u \in U$ and $w \in W$, this means that $u \in U \cap W$ and $w = -u$.

Therefore $\ker(\varphi) = \{(z, -z) \mid z \in U \cap W\}$.

The map $z \mapsto (z, -z)$ is an isomorphism from $U \cap W$ to $\ker(\varphi)$. Hence $\dim(\ker(\varphi)) = \dim(U \cap W)$.

Rank-nullity. By rank-nullity applied to $\varphi : U \times W \rightarrow U + W$, we get $\text{rank}(\varphi) + \dim(\ker(\varphi)) = \dim(U \times W)$.

Using the results above, this becomes $\dim(U + W) + \dim(U \cap W) = \dim(U \times W)$.

Finally, by the dimension formula for product spaces, $\dim(U \times W) = \dim(U) + \dim(W)$.

Thus $\dim(U + W) + \dim(U \cap W) = \dim(U) + \dim(W)$, and therefore $\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$. $\square$

Corollary
If $U \oplus W$ is a direct sum, then $\dim(U \oplus W) = \dim(U) + \dim(W).$

Proof. If $U \oplus W$ is a direct sum, then $U \cap W = \{\mathbf{0}\}$. Therefore $\dim(U \cap W) = 0$.

Applying the previous proposition gives $\dim(U + W) = \dim(U) + \dim(W) - 0$.

Since in the direct-sum case we write $U + W = U \oplus W$, we obtain $\dim(U \oplus W) = \dim(U) + \dim(W)$. $\square$